Year 1 AS Pure Maths

Chapter 9

Trigonometric ratios

Practice Booklet

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9.1 The cosine rule

Practice Book, printed page 68–69.

1. In triangle ABC, $AB = 7.2$ cm, $BC = 9.3$ cm and $\angle ABC = 110°$. Find the length of side AC.

Hint

Use this version of the cosine rule if you know two sides and the angle between them and want to find the missing side:

$$a^2 = b^2 + c^2 - 2bc\cos A$$

2. Find the size of the largest angle in a triangle with sides of length 5 cm, 6.5 cm and 10 cm.

Hint

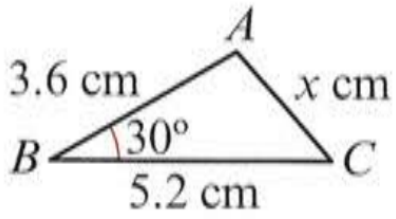
Use this version of the cosine rule to find an angle if you know all three sides:

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

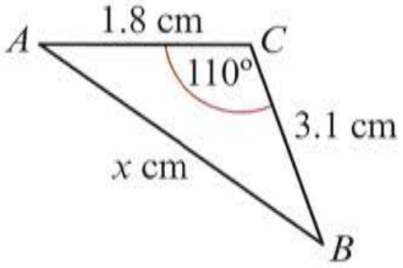
In any triangle, the largest angle is opposite the longest side.

3. In each of these triangles calculate the value of x. Give your answers to 3 significant figures.

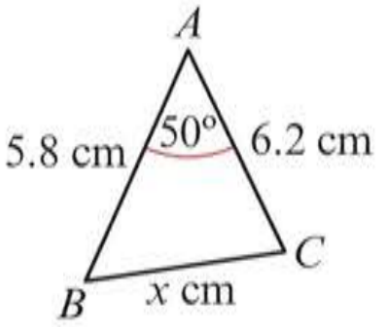
a



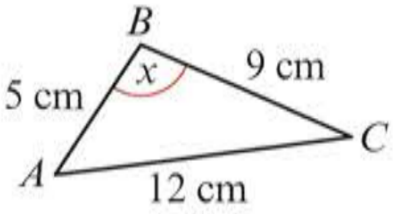
b



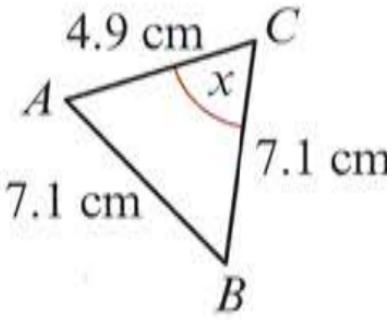
c



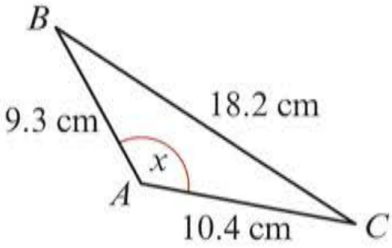
d



e



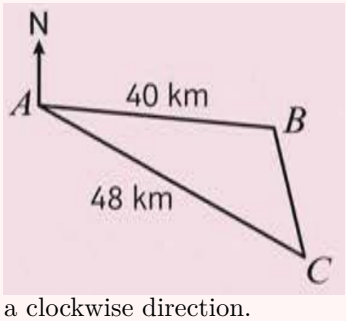
f



4. Three towns A, B and C are positioned such that:
B is 40 km from A on a bearing of 105°
C is 48 km from A on a bearing of 135°
Calculate the distance between B and C.

Hint

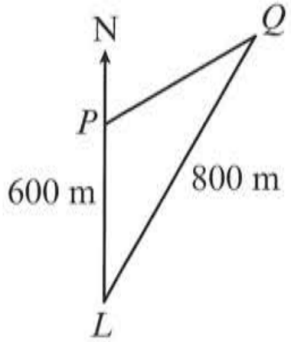
Draw your own clearly labelled diagram in questions where one is not provided.



Bearings are measured from north in a clockwise direction.

5. E In triangle ABC, $AB = 5$ cm, $BC = 8$ cm, $AC = 6$ cm and $\angle BAC = \theta$.
Show that $\cos\theta = -\frac{1}{20}$. (2 marks)

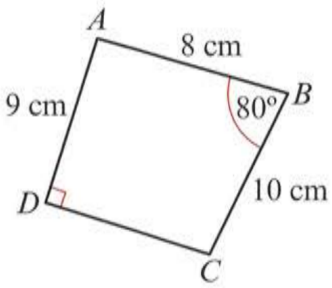
6. E The diagram shows a lighthouse L and two boats P and Q which are assumed to be in the same horizontal plane.
Boat P is 600 m due north of L and boat Q is 800 m from L on a bearing of 020°.



Calculate the distance between boat P and boat Q. (3 marks)

7. E/P In triangle ABC, $AB = 7$ cm, $BC = 12$ cm and $AC = 10$ cm.
Use the cosine rule to show that triangle ABC does **not** contain an obtuse angle. (3 marks)

8. E/P $ABCD$ is a quadrilateral. $AB = 8$ cm, $BC = 10$ cm and $AD = 9$ cm.
$\angle ABC = 80°$ and $\angle ADC = 90°$.



Calculate the perimeter of $ABCD$. (5 marks)

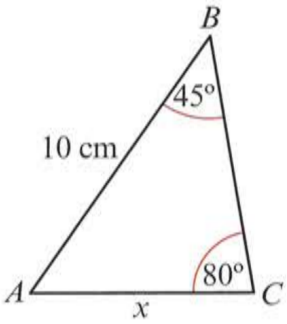
9. E/P In triangle ABC, $AB = (x + 2)$ cm, $AC = (x + 1)$ cm and $BC = x$ cm.
Given that $\cos \angle BAC = \frac{3}{4}$, find the value of x. (6 marks)

9.2 The sine rule

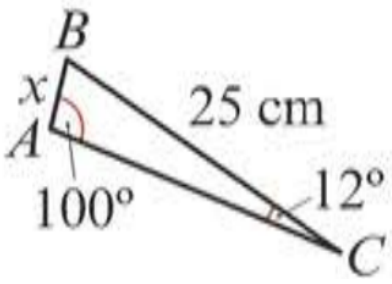
Practice Book, printed page 69–70.

1. Work out the length of the side marked x in each triangle:

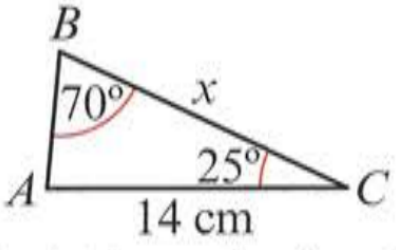
a



b



c



Hint

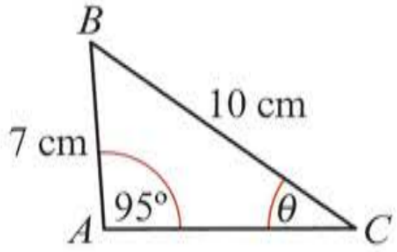
Use this version of the sine rule to find the length of a missing side:

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

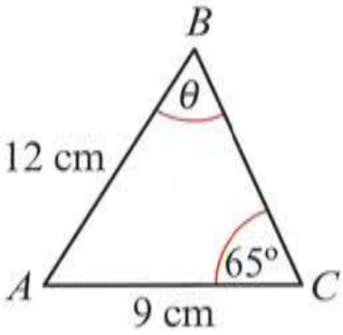
If you are given two angles in a triangle you can find the third angle using
Sum of angles in a triangle = 180°

2. Work out the size of the angle marked θ in each triangle:

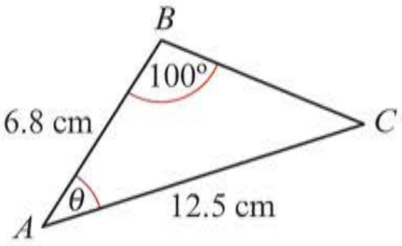
a



b



c

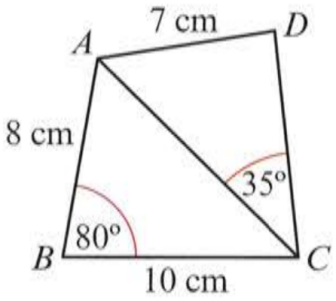


Hint

Use this version of the sine rule to find a missing angle:

$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$

3. In the diagram find:



a. $\angle ADC$

b. the length of CD.

Hint

Start by using trigonometry on triangle ABC to find the length of AC.

4. In triangle ABC, $AB = 6$ cm, $AC = 4$ cm and $\angle ABC = 40°$.

a. Calculate the two possible values of $\angle ACB$.

b. Draw a diagram to illustrate your answers.

Hint

As $\sin\theta = \sin(180° - \theta)$, the sine rule sometimes produces two possible solutions for a missing angle. You need to check the sum of the angles of the triangle to see if there are two possible solutions.

5. E In triangle ABC, $AC = 18$ cm, $BC = 20$ cm and $\angle BAC = 50°$.
Calculate the size of $\angle ACB$. (3 marks)

6. E/P In triangle ABC, $AB = 9$ cm, $AC = 8$ cm, $\angle ABC = 25°$ and $\angle BCA = x$.

a. Find the value of $\sin x$, giving your answer to 3 decimal places. (2 marks)

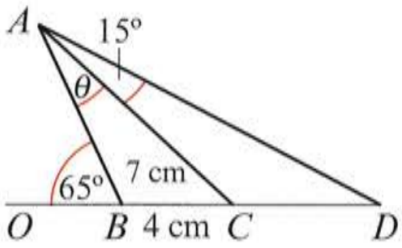
b. Given that there are two possible values of x, find these values of x, giving your answers to 2 decimal places. (3 marks)

7. E/P Three supermarkets A, B and C lie in the same horizontal plane and are positioned such that:
B is 6 km from A on a bearing of 015°
C is on a bearing of 045° from A
The distance between B and C is 14 km. Calculate:

a. the distance of C from A (3 marks)

b. the bearing of C from B to the nearest degree. (4 marks)

8. E/P In the diagram, $AC = 7$ cm, $BC = 4$ cm, $\angle OBA = 65°$, $\angle CAD = 15°$ and $\angle BAC = \theta$.



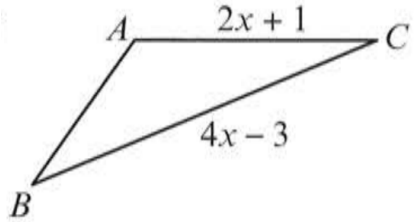
Find:

a. the size of angle θ (4 marks)

b. the length of AD (4 marks)

c. the length of CD. (3 marks)

9. E/P In the diagram, $AC = 2x + 1$, $BC = 4x - 3$, $\sin A = \frac{\sqrt{3}}{2}$ and $\sin B = \frac{3}{4}$.



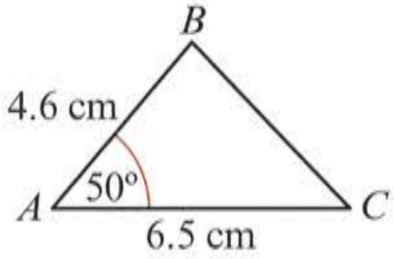
Show that the exact value of x is $\frac{11+5\sqrt{3}}{8}$. (7 marks)

9.3 Areas of triangles

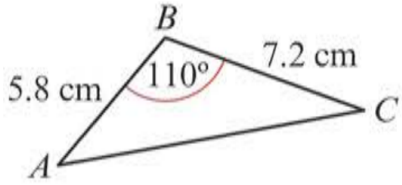
Practice Book, printed page 71–72.

1. Calculate the area of each triangle.

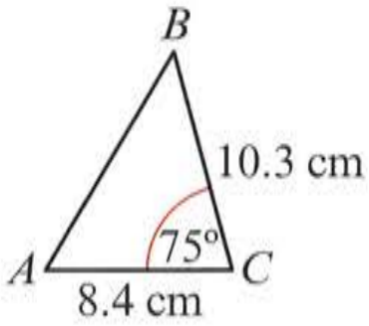
a



b



c



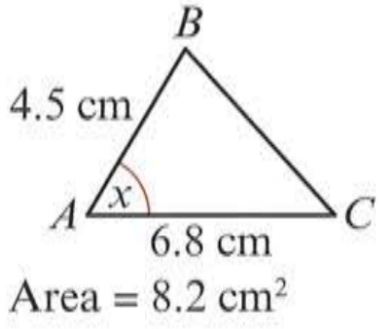
Hint

You can find the area of a triangle when you know two sides and the angle between them.

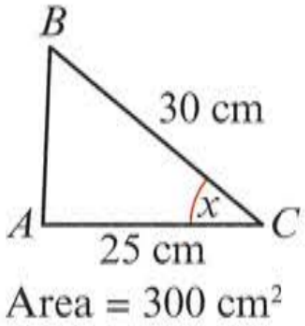
$$\text{Area} = \tfrac{1}{2}ab\sin C$$

2. Work out the possible sizes of angle x in each triangle.

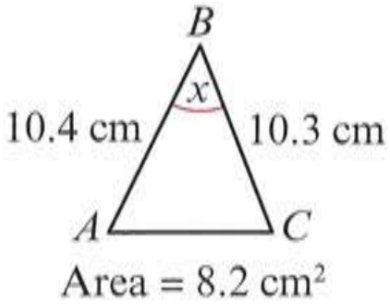
a



b



c



Hint

There are two possible solutions for each triangle. Use $\sin\theta = \sin(180° - \theta)$ to find one acute angle and one obtuse angle.

3. The area of a triangular plot of land is 1500 m^2. The lengths of two sides of the triangle are 75 m and 80 m, and the angle between them is θ.

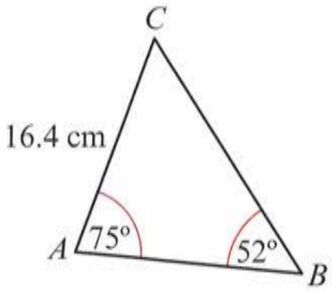
a. Show that a possible size of angle θ is 150°.

b. Work out the total perimeter of the plot of land.

Hint

For part **b**, use the cosine rule to work out the length of the longest side, as you know two sides and the angle between them.

4. E The diagram shows a triangle ABC. The length of AC is 16.4 cm.
$\angle BAC = 75°$ and $\angle ABC = 52°$.



a. Find the length of BC to the nearest 0.1 cm. (3 marks)

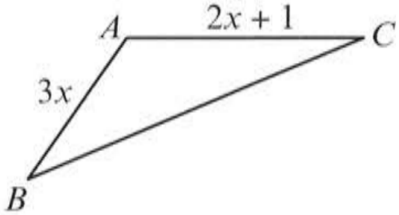
b. Calculate the area of triangle ABC, giving your answer to the nearest cm^2. (3 marks)

5. E/P In triangle PQR, $PR = x + 3$, $QR = 6 - x$ and $\sin\angle PRQ = \frac{2}{5}$.
The area of the triangle is A cm^2.

a. Find an expression for A in the form $\frac{ax^2+bx+c}{d}$ where a, b, c and d are integers. (3 marks)

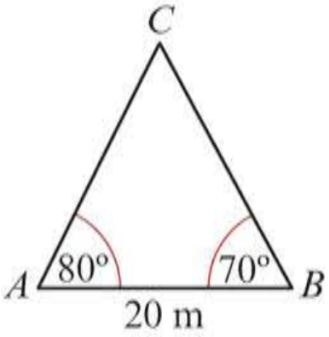
b. By completing the square or otherwise, find the maximum value of A and state the corresponding value of x. (4 marks)

6. E/P In triangle ABC, $AB = 3x$ and $AC = 2x + 1$.



Given that $\sin\angle BAC = \frac{3}{8}$ and the area of the triangle is 43 cm^2, calculate the value of x, giving your answer to 3 significant figures. (5 marks)

7. E/P A triangular floor space is modelled by the triangle ABC shown in the diagram. The length of AB is 20 m, $\angle BAC = 80°$ and $\angle ABC = 70°$.



a. Calculate the area of the floor space to 3 significant figures. (4 marks)

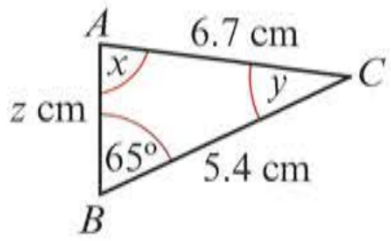
b. Why is your answer unlikely to be accurate to the nearest square metre? (1 mark)

9.4 Solving triangle problems

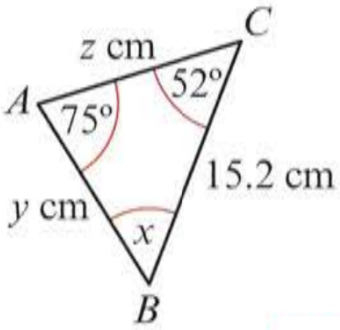
Practice Book, printed page 72–73.

1. In each triangle find the values of x, y and z. Give your answers to 3 significant figures.

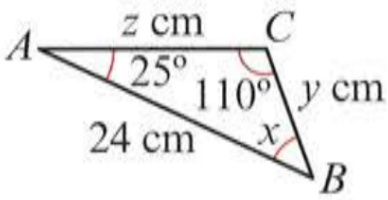
a



b

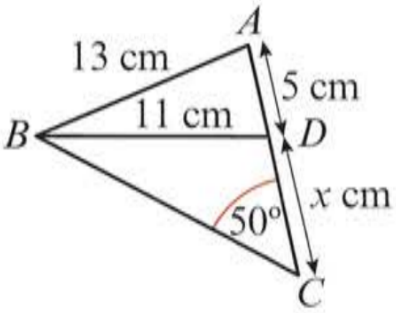


c

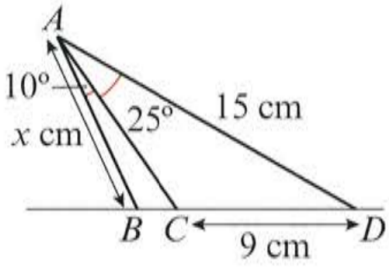


2. In each triangle find the value of x. Give your answers to 3 significant figures.

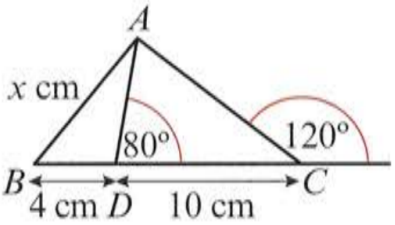
a



b



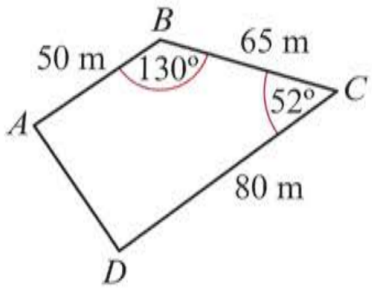
c



Hint

If you have a choice of method, the sine rule is often easier to use than the cosine rule.

3. The diagram shows the dimensions of a field. $AB = 50$ m, $BC = 65$ m and $CD = 80$ m. $\angle ABC = 130°$ and $\angle BCD = 52°$.

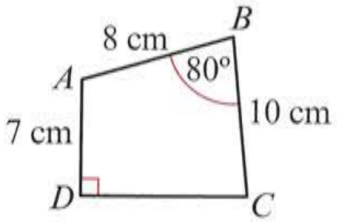


Calculate the area of the field.

Hint

Split the quadrilateral into two triangles, ABC and ACD.

4. E/P $ABCD$ is a quadrilateral. $AB = 8$ cm, $AD = 7$ cm and $BC = 10$ cm. $\angle ABC = 80°$ and $\angle ADC = 90°$.



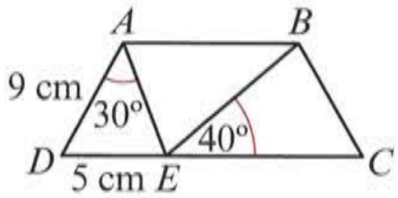
Calculate the area of quadrilateral $ABCD$. (7 marks)

5. E/P In triangle ABC, $AB = 15$ cm, $BC = 8$ cm and $\angle ABC = \theta$. The area of triangle ABC is 48 cm^2.

a. Find the two possible values of $\cos\theta$. (4 marks)

b. Given that AC is the longest side of the triangle, find the exact length of AC. (2 marks)

6. E/P The diagram shows a trapezium $ABCD$. $AD = 9$ cm and $DE = 5$ cm. $\angle DAE = 30°$ and $\angle BEC = 40°$.



Find the area of triangle ABE. (7 marks)

7. E/P A ship A is 10 km from a lighthouse L on a bearing of 080°. A ship B is 6.5 km from L on a bearing of 200°.

a. Find the distance between A and B. (5 marks)

b. Find the bearing of A from B to the nearest degree. (5 marks)

9.5 Graphs of sine, cosine and tangent

Practice Book, printed page 73–74.

1.

a. Sketch the graph of $y = \sin\theta$ for $-360° \leqslant \theta \leqslant 360°$.

b. State the period of the graph of $y = \sin\theta$.

c. State the maximum and minimum values of y.

Hint

The period of a graph is the interval covered on the horizontal axis before the graph starts to repeat itself.

2.

a. Sketch the graph of $y = \cos\theta$ for $-360° \leqslant \theta \leqslant 360°$.

b. State the period of the graph of $y = \cos\theta$.

c. State the maximum and minimum values of y.

Hint

The graph of $y = \cos\theta$ has the same shape as the graph of $y = \sin\theta$, but translated horizontally.

3.

a. Sketch the graph of $y = \tan\theta$ for $-360° \leqslant \theta \leqslant 360°$.

b. State the period of the graph of $y = \tan\theta$.

Hint

The graph of $y = \tan\theta$ has no maximum or minimum values.
It has vertical asymptotes at intervals at $\theta = \ldots, -270°, -90°, 90°, 270°, \ldots$

4. Use your answers to questions **1**–**3** to identify, for $-360° \leqslant \theta \leqslant 360°$, the values of θ for which:

a. $\sin\theta = 1$

b. $\cos\theta = -1$

c. $\tan\theta = 0$

5. E/P

a. Use a calculator to find a solution to $\sin\theta = 0.5$. (1 mark)

b. Sketch the graphs of $y = \sin\theta$ and $y = 0.5$ for $-360° \leqslant \theta \leqslant 360°$ on the same set of axes. (3 marks)

c. Use your graphs to solve the equation $\sin\theta = 0.5$ for $-360° \leqslant \theta \leqslant 360°$. (2 marks)

6. E/P

a. Use a calculator to find a solution to $\tan\theta = -1$. (1 mark)

b. Sketch the graphs of $y = \tan\theta$ and $y = -1$ for $-360° \leqslant \theta \leqslant 360°$ on the same set of axes. (3 marks)

c. Use your graphs to solve the equation $\tan\theta = -1$ for $-360° \leqslant \theta \leqslant 360°$. (2 marks)

7. E/P

a. Sketch the graph of $y = \cos\theta$ for $-360° \leqslant \theta \leqslant 360°$. (2 marks)

b. Given $\cos 45° = \frac{1}{\sqrt{2}}$, use your graph to find all solutions for $-360° \leqslant \theta \leqslant 360°$ to:

i. $\cos\theta = \frac{1}{\sqrt{2}}$

ii. $\cos\theta = -\frac{1}{\sqrt{2}}$

(4 marks)

9.6 Transforming trigonometric graphs

Practice Book, printed page 74–75.

1.

a. On the same set of axes, sketch the graphs of $y = \cos x$, $y = 2\cos x$ and $y = -\cos x$ for $-360° \leqslant x \leqslant 360°$.

b. Describe the transformation which maps the graph of $y = \cos x$ onto the graph of:

i. $y = 2\cos x$

ii. $y = -\cos x$

Hint

A transformation from the graph of $y = f(x)$ onto the graph of $y = af(x)$ is a **stretch** by scale factor a in the vertical direction.
The graph of $y = -f(x)$ is a **reflection** of the graph of $y = f(x)$ in the x-axis. ← Section 4.6

2.

a. On the same set of axes, sketch the graphs of $y = \sin x$ and $y = \sin 3x$ for $0 \leqslant x \leqslant 360°$.

b. On a second set of axes, sketch the graphs of $y = \tan x$ and $y = \tan(-x)$ for $-180° \leqslant x \leqslant 180°$.

c. Describe the transformation which maps the graph of:

i. $y = \sin x$ onto the graph of $y = \sin 3x$

ii. $y = \tan x$ onto the graph of $y = \tan(-x)$

Hint

A transformation from the graph of $y = f(x)$ onto the graph of $y = f(ax)$ is a **stretch** by scale factor $\frac{1}{a}$ in the horizontal direction.
The graph of $y = f(-x)$ is a **reflection** of the graph of $y = f(x)$ in the y-axis. ← Section 4.6

3.

a. On the same set of axes, sketch the graphs of $y = \sin x$ and $y = \sin x + 1$ for $0 \leqslant x \leqslant 360°$.

b. Describe the transformation which maps the graph of $y = \sin x$ onto the graph of $y = \sin x + 1$.

Hint

A transformation from the graph of $y = f(x)$ onto the graph of $y = f(x) + a$ is a **translation** by vector $\begin{pmatrix} 0 \\ a \end{pmatrix}$. ← Section 4.5

4.

a. On the same set of axes, sketch the graphs of $y = \tan x$ and $y = \tan(x + 30°)$ for $0 \leqslant x \leqslant 360°$.

b. Describe the transformation which maps the graph of $y = \tan x$ onto the graph of $y = \tan(x + 30°)$.

Hint

A transformation from the graph of $y = f(x)$ onto the graph of $y = f(x + a)$ is a **translation** in the x-axis by vector $\begin{pmatrix} -a \\ 0 \end{pmatrix}$. ← Section 4.5

5. E/P

a. On the same set of axes, sketch the graphs of $y = \cos x$ and $y = \cos 2x$ for $0 \leqslant x \leqslant 360°$. (3 marks)

b. Describe the transformation which maps the graph of $y = \cos x$ onto the graph of $y = \cos 2x$. (2 marks)

c. State the period of the graph of $y = \cos 2x$. (1 mark)

d. Write down the values of x for which $\cos 2x = 0$ in the interval $0 \leqslant x \leqslant 360°$. (2 marks)

6. E/P

a. On the same set of axes, sketch the graphs of $y = \sin x$ and $y = \sin(x - 20°)$ for $0 \leqslant x \leqslant 360°$. (3 marks)

b. Describe the transformation which maps the graph of $y = \sin x$ onto the graph of $y = \sin(x - 20°)$. (2 marks)

c. Identify the value of x in the interval $0 \leqslant x \leqslant 360°$ for which:

i. $\sin(x - 20°) = 1$

ii. $\sin(x - 20°) = -1$

(2 marks)

7. E/P

a. On the same set of axes, sketch the graphs of $y = \cos x$ and $y = 5\cos x$ for $-180° \leqslant x \leqslant 180°$. (3 marks)

b. Describe the transformation which maps the graph of $y = \cos x$ onto the graph of $y = 5\cos x$. (2 marks)

c. Identify the maximum and minimum values of $5\cos x$. (1 mark)

d. Given that the equation $5\cos x = k$, for $-180° \leqslant x \leqslant 180°$, where k is constant, has one solution $x = \theta$, state, in terms of θ, the second solution. (1 mark)

8. E/P

a. Sketch the graphs of $y = \sin x$ and $y = \cos x$ on the same set of axes for $0 \leqslant x \leqslant 360°$. (3 marks)

b. Use your graphs to identify the number of solutions to the equation $\sin x = \cos x$ for $0 \leqslant x \leqslant 360°$. (1 mark)

c. State the solutions to the equation $\sin x = \cos x$, for $0 \leqslant x \leqslant 360°$. (2 marks)

d. Find one possible value for k, where k is a constant, for which:

i. $\sin x = \cos(x + k)$

ii. $\cos x = \sin(x + k)$

(2 marks)

Problem set A

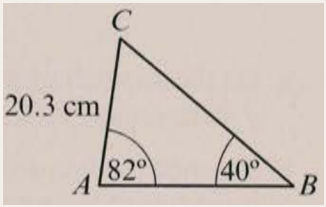
Bronze

The diagram shows a triangle ABC.
The length of AC is 20.3 cm, $\angle BAC = 82°$ and $\angle ABC = 40°$.
Calculate:

a. the length of BC correct to the nearest 0.1 cm (3 marks)

b. the area of triangle ABC, giving your answer to the nearest cm^2 (3 marks)

c. the perimeter of triangle ABC, giving your answer to the nearest 0.1 cm. (3 marks)



Silver

A hiker walking due north on a straight path sees a wind turbine, W, on a bearing of 040°. After walking 500 m, the bearing of W from the hiker's new position is 075°. Find:

a. the distance between W and the hiker's new position (5 marks)

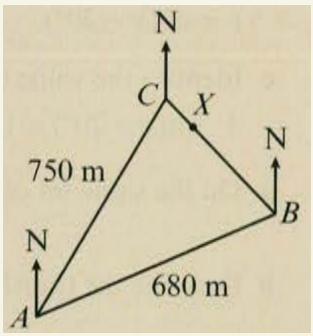
b. the distance of W from the path. (3 marks)

Gold

A man walks 750 m from a point A to a point C on a bearing of 042°.
He then walks along a straight path from C on a bearing of 162° towards a point B.
When the man reaches a point X on the path, his distance from point A is the same as the distance between point B and point A.
Given that the distance $AB = 680$ m, calculate:

a. the distance XB (7 marks)

b. the bearing of A from B. (3 marks)



Problem set B

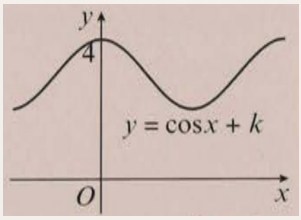
Bronze

The graph shows the curve with equation $y = \cos x + k$, where k is a constant.
The graph passes through the point with coordinates $(0, 4)$.

a. Find the value of k. (2 marks)

b. Use your value of k to find the coordinates of the minimum points for the graph of $y = \cos x + k$ for $-180° \leqslant x \leqslant 360°$. (2 marks)

c. Find, in terms of A, the coordinates of the point where the graph with equation $y = \cos x + A$ crosses the y-axis. (2 marks)



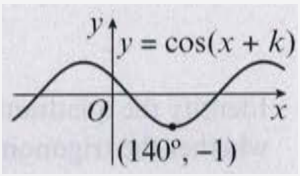
Silver

The graph shows the curve with equation $y = \cos(x + k)$ where k is a positive constant.
The graph passes through the point with coordinates $(140°, -1)$.

a. State the smallest possible value of k. (2 marks)

b. Use your value of k to find the solutions to the equation $\cos(x + k) = 0$ for $-180° \leqslant x \leqslant 360°$. (2 marks)

c. Identify another possible value for k for the curve. Explain your answer. (2 marks)



Gold

$f(x) = \cos kx$, $0 \leqslant k \leqslant 360°$, where k is a positive constant.

a. In the case when k is a positive integer, write, in terms of k, the number of solutions to the equation $f(x) = 0.5$. (2 marks)

b. In the case when $k = 1.5$, sketch the graph of $y = f(x)$, showing clearly any points where the curve cuts the coordinate axes. (2 marks)

c. Given that the equation $f(x) = 0.5$ has exactly three real roots, state the range of possible values of k. (3 marks)

Selected exam question bank questions

Question 17

The graph of $y = \tan(x + k)$ passes through the point $(60°, 1)$.

a. Find one positive and one negative possible value of k. (3 marks)

b. Give the equations of two different asymptotes. (3 marks)

Question 31

A parallelogram has sides of lengths 9 cm and 7.5 cm.

The size of the larger interior angles is θ.

Given that the area of the parallelogram is 60 cm^2, find the exact value of $\tan\theta$. (4 marks)

Question 37

In the triangle ABC, $AB = 14$ cm, $BC = 6$ cm and $\angle ABC = \theta$. The area of the triangle is 24 cm^2.

a. Find the two possible values of θ correct to 1 decimal place. (3 marks)

b. Given that AC is the longest side of the triangle, find the length of AC. (2 marks)

Question 39

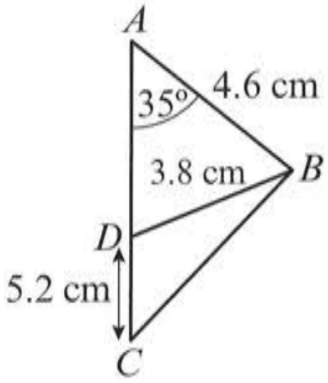
a. On the same set of axes, sketch the graphs of $y = \cos x$ and $y = \cos 3x$ for $0 \leqslant x \leqslant 360°$. (3 marks)

b. Describe the transformation from the graph of $y = \cos x$ to $y = \cos 3x$. (2 marks)

c. Given that $\cos 3x = \cos 60°$, use your graph to find all the solutions to the equation $\cos 3x = \cos 60°$ for $0 \leqslant x \leqslant 360°$. (4 marks)

Question 45

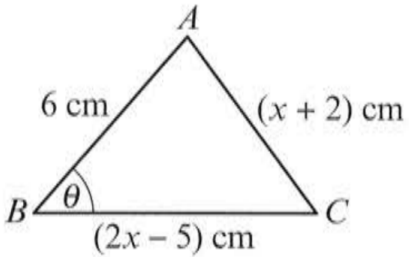
The diagram shows triangle ABC.



Calculate the area of triangle ABC. (6 marks)

Question 63

The diagram shows triangle ABC, with $AB = 6$ cm, $BC = (2x - 5)$ cm and $AC = (x + 2)$ cm.



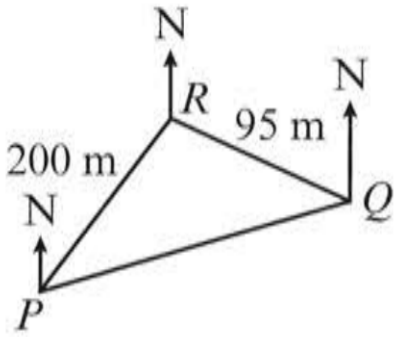
Given that $\angle ABC = \theta$ where $\cos\theta = \frac{3}{4}$,

a. show that $x^2 - 14x + 34 = 0$ (3 marks)

b. By completing the square or otherwise, find the exact values of the two possible values of x. (3 marks)

Question 73

The diagram shows three points P, Q and R in the same horizontal plane.
R is 200 m from P on a bearing of 068°. R is 95 m from Q on a bearing of 310°.



Calculate:

a. the distance PQ (5 marks)

b. the bearing of Q from P. (5 marks)

Question 78

a. On the same set of axes sketch the graphs of $y = \cos(x - 60°)$ and $y = -\cos x$, in the interval $-90° \leqslant x \leqslant 360°$. (5 marks)

b. State the coordinates of the points of intersection of the graph of $y = \cos(x - 60°)$ with the x- and y-axes for $-90° \leqslant x \leqslant 360°$. (3 marks)

c. Use your graph to find the solutions to the equation $\cos(x - 60°) + \cos x = 0$ in the interval $-90° \leqslant x \leqslant 360°$. (3 marks)

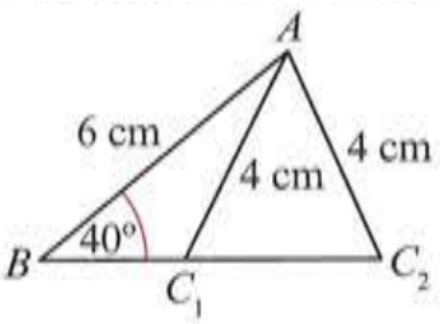
Answers

9.1 The cosine rule

1. 13.6 cm (3 s.f.)
2. 120° (3 s.f.)
3.
   a. 2.75 (3 s.f.)
   b. 4.08 (3 s.f.)
   c. 5.08 (3 s.f.)
   d. 115 (3 s.f.)
   e. 69.8 (3 s.f.)
   f. 135 (3 s.f.)
4. 24.1 km (3 s.f.)
5. $\cos\theta = \dfrac{5^2 + 6^2 - 8^2}{2 \times 5 \times 6} = -\dfrac{1}{20}$
6. 313 m (3 s.f.)
7. Largest angle is $\angle BAC$.
   $\cos A = \dfrac{10^2 + 7^2 - 12^2}{2 \times 10 \times 7} = \dfrac{5}{140} > 0$
   So triangle ABC does **not** contain an obtuse angle.
8. 34.4 cm (3 s.f.)
9. 4

9.2 The sine rule

1.
   a. 7.18 cm (3 s.f.)
   b. 5.28 cm (3 s.f.)
   c. 14.8 cm (3 s.f.)
2.
   a. 44.2° (3 s.f.)
   b. 42.8° (3 s.f.)
   c. 47.6° (3 s.f.)
3.
   a. 73.0° (3 s.f.)
   b. 11.6 cm (3 s.f.)
4.
   a. 74.6° and 105.4° (1 d.p.)
   b.



5. 86.4° (1 d.p.)
6.
   a. 0.475
   b. 28.39°, 151.61°

7.
- **a.** 18.9 km (3 s.f.)
- **b.** 057°

8.
- **a.** 31.2° (3 s.f.)
- **b.** 12.1 cm (3 s.f.)
- **c.** 5.62 cm (3 s.f.)

9. Using the sine rule, $\frac{2(4x-3)}{\sqrt{3}} = \frac{4(2x+1)}{3}$
Rearrange for x and rationalise the denominator.

9.3 Areas of triangles

1.
- **a.** 11.5 cm^2 (3 s.f.)
- **b.** 19.6 cm^2 (3 s.f.)
- **c.** 41.8 cm^2 (3 s.f.)

2.
- **a.** 32.4° and 148° (3 s.f.)
- **b.** 53.1° and 127° (3 s.f.)
- **c.** 8.81° and 171° (3 s.f.)

3.
- **a.** $1500 = \frac{1}{2} \times 75 \times 80 \times \sin\theta \Rightarrow \sin\theta = \frac{1}{2}$
$\Rightarrow \theta = 30°$ or 150°
- **b.** Perimeter = 305 m (3 s.f.)

4.
- **a.** 20.1 cm
- **b.** 132 cm^2

5.
- **a.** $A = \frac{18+3x-x^2}{5}$ cm^2
- **b.** 4.05 cm^2, when $x = 1.5$ cm

6. 5.94 cm

7.
- **a.** 370 m^2
- **b.** Because the angles and side length are not given to 4 significant figures.

9.4 Solving triangle problems

1.
- **a.** $x = 46.9°$, $y = 68.1°$, $z = 6.86$
- **b.** $x = 53.0°$, $y = 12.4$, $z = 12.6$
- **c.** $x = 45.0°$, $y = 10.8$, $z = 18.1$

2.
- **a.** $x = 11.3$
- **b.** $x = 6.21$
- **c.** $x = 14.7$

3. 3360 m^2 (3 s.f.)

4. 72.1 cm^2 (3 s.f.)

5.
- **a.** $\pm\frac{3}{5}$
- **b.** $\sqrt{433}$ cm

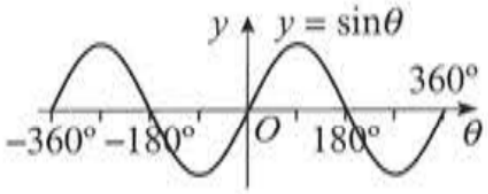
6. 67.5 cm^2

7.
- **a.** 14.4 km (3 s.f.)
- **b.** 057°

9.5 Graphs of sine, cosine and tangent

1.

a.

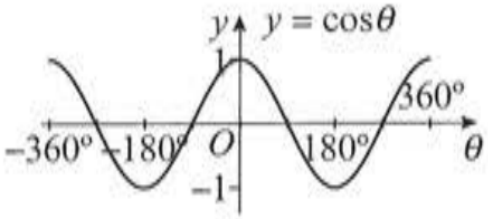


b. 360°

c. maximum 1, minimum −1

2.

a.

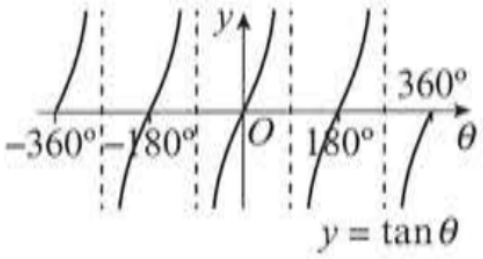


b. 360°

c. maximum 1, minimum −1

3.

a.



b. 180°

4.

a. −270° and 90°

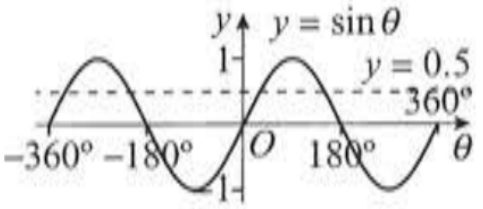
b. −180° and 180°

c. −360°, −180°, 0°, 180° and 360°

5.

a. $\theta = 30°$

b.

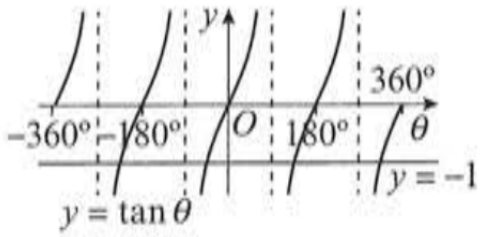


c. $\theta = -330°, -210°, 30°, 150°$

6.

a. $-45°$

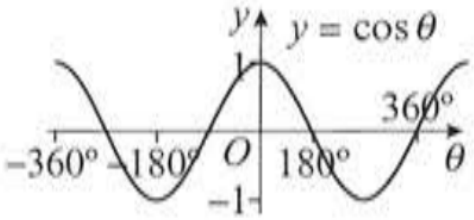
b.



c. $\theta = -225°, -45°, 135°, 315°$

7.

a.



b.

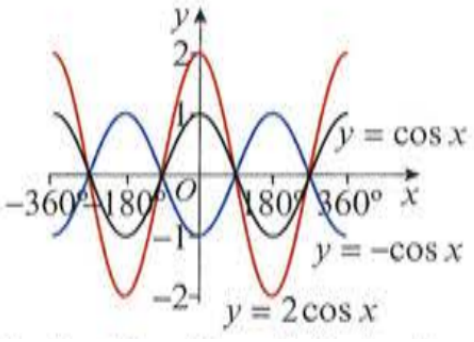
i. $\theta = \pm 315°, \pm 45°$

ii. $\theta = \pm 225°, \pm 135°$

9.6 Transforming trigonometric graphs

1.

a.

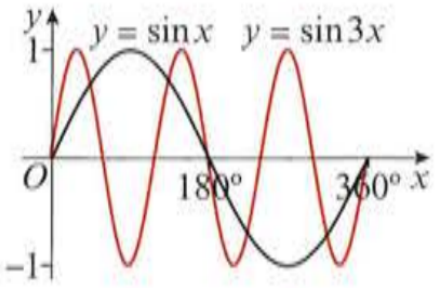


b.

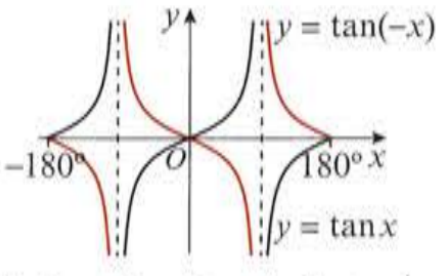
i. Stretch with scale factor 2 parallel to the y-axis

ii. Reflection in the x-axis

2. **a.**



b.

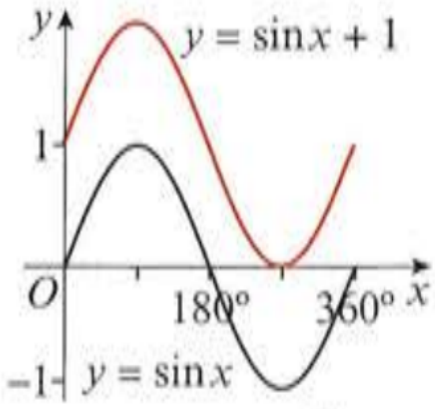


c.

i. Stretch with scale factor $\frac{1}{3}$ parallel to the x-axis

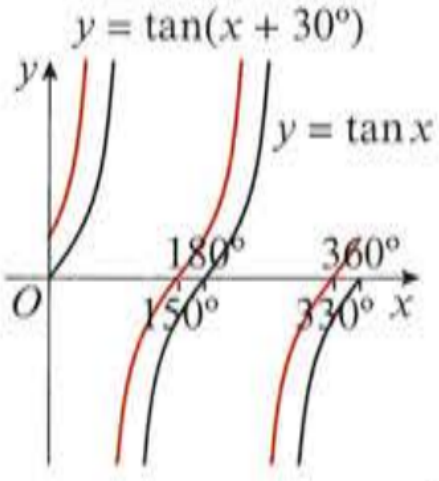
ii. Reflection in the y-axis

3. **a.**



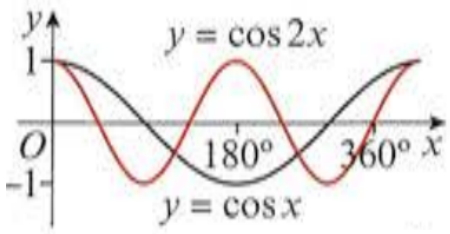
b. Translation by $\begin{pmatrix}0\\1\end{pmatrix}$

4. **a.**



b. Translation by vector $\begin{pmatrix}-30°\\0\end{pmatrix}$

5. **a.**

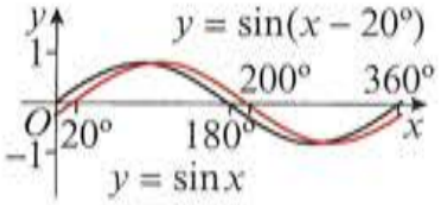


b. Stretch with scale factor $\frac{1}{2}$ parallel to the x-axis

c. 180°

d. 45°, 135°, 225°, 315°

6. **a.**



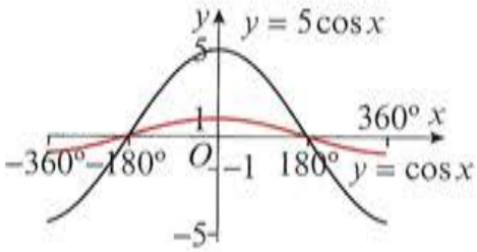
b. Translation by vector $\begin{pmatrix}20^\circ\\0\end{pmatrix}$

c.

i. 110°

ii. 290°

7. **a.**

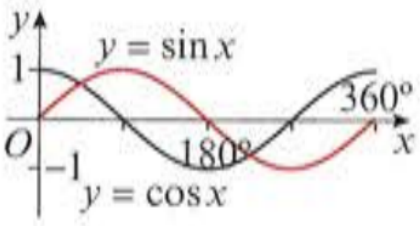


b. Stretch with scale factor 5 parallel to the y-axis.

c. maximum 5, minimum -5

d. $-\theta$

8. **a.**



b. 2

c. $x = 45^\circ$ and $x = 225^\circ$

d.

i. e.g. $k = -90^\circ$

ii. e.g. $k = 90^\circ$

Problem set A

Bronze

a. 31.3 cm
b. 269 cm^2
c. 78.4 cm

Silver

a. 560 m (3 s.f.)
b. 541 m (3 s.f.)

Gold

a. 403 m (3 s.f.)
b. 269° (nearest degree)

Problem set B

Bronze

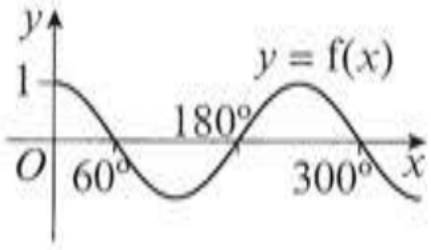
a. 3
b. $(-180°, 2)$ and $(180°, 2)$
c. $(0, A + 1)$

Silver

a. $k = 40$
b. $x = -130°, 50°, 230°$
c. E.g. $k = 400$. As the period of the graph is 360°, k can take any value of the form $40 \pm 360n$

Gold

a. $2k$
b.



c. $\frac{7}{6} \leqslant k < \frac{11}{6}$

Selected exam question bank answers

Question 17

a. e.g. $k = -15°$, $k = 165°$
b. e.g. $x = 105°$, $x = -75°$

Question 31 $-\frac{8}{\sqrt{17}}$

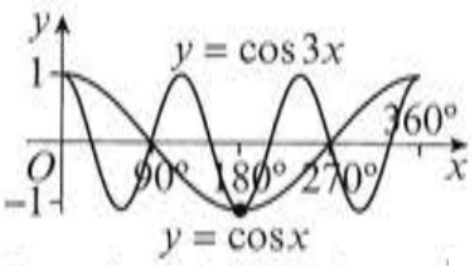
Question 37

a. 34.8° and 145.2° (1 d.p.)

b. 19.2 cm (3 s.f.)

Question 39

a.



b. Stretch with scale factor $\frac{1}{3}$ parallel to the x-axis.

c. $x = 20°, 100°, 140°, 220°, 260°$ or $340°$

Question 45 15.4 cm^2 (3 s.f.)

Question 63

a. Using cosine rule, $\frac{3}{4} = \dfrac{6^2 + (2x-5)^2 - (x+2)^2}{2 \times 6 \times (2x-5)}$

b. $x = 7 \pm \sqrt{15}$

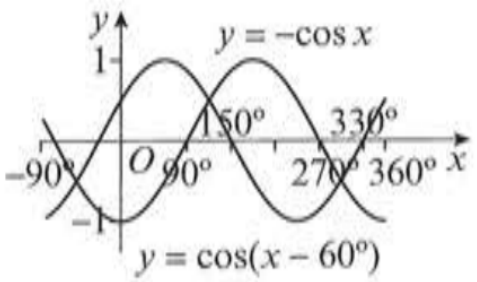
Question 73

a. 258.6 m (1 d.p.)

b. 087° (nearest degree)

Question 78

a.



b. $(0, 0.5)$, $(-30°, 0)$, $(150°, 0)$, $(330°, 0)$

c. $x = -60°$, $120°$ or $300°$